

Entropy Decay Mechanism and the Honest Boundary of #21

$dH/dt = -\frac{\epsilon_0^2}{2}(1 - \frac{1}{n})\sum_k P_k^{-1}$ Derived from First Principles
What closes and what remains for the bracket identity $\kappa/\ln 2$

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Abstract

This document derives the entropy decay mechanism between R-events from the δ -chain without new axioms, and gives the honest boundary of Instability #21.

New derivation: between R-events, P_i evolves only via ϵ_0 background noise. A second-order Taylor expansion of $H(P + \epsilon_0\xi)$ gives:

$$\left.\frac{dH}{dt}\right|_{\text{noise}} = -\frac{\epsilon_0^2}{2}\left(1 - \frac{1}{n}\right)\sum_{k=1}^n \frac{1}{P_k} < 0$$

This is the mechanism behind $H_{\text{before}} < H_{\text{after}}$. Verified numerically within factor 2 of the measured decay rate (discrepancy from not knowing the exact P distribution at R-events).

The final gap for Instability #21: the bracket identity $\text{bracket} = \kappa/\ln 2$ requires knowing $\sum_k P_k^{-1}$ at the moment of R-events, which requires the P distribution under combined ϵ_0 noise and Born-rule dynamics. Fastest path: instrument V20.3 to directly record $\sum_k P_k^{-1}$ at each event.

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1 What Updates P_i Between R-Events

From V20.3 source code (line 294):

$$P_i \leftarrow P_i + \epsilon_0 \xi_i, \quad \xi_i \sim \mathcal{N}(0, I_n), \quad \text{then renormalised}$$

Phases Φ and positions X update separately but do *not* affect P_i . Between Born-rule resets, P_i evolves *only* via ϵ_0 background noise. This is the sole source of entropy change between R-events.

2 Deriving the Entropy Decay Rate

Derived: Entropy ODE from ϵ_0 Noise

Let $H = -\sum_k P_k \ln P_k$. After one sweep of noise $\delta P_k = \epsilon_0 \xi_k - \epsilon_0 \bar{\xi}$ (renormalised), with $\text{Cov}[\delta P_k, \delta P_l] = \epsilon_0^2 (\delta_{kl} - 1/n)$:

$$\mathbb{E}[\delta H] = \underbrace{\sum_k \frac{\partial H}{\partial P_k} \mathbb{E}[\delta P_k]}_{=0} + \frac{1}{2} \sum_{k,l} \frac{\partial^2 H}{\partial P_k \partial P_l} \text{Cov}[\delta P_k, \delta P_l]$$

Since $\partial^2 H / \partial P_k^2 = -1/P_k$ and $\partial^2 H / \partial P_k \partial P_l = 0$ ($k \neq l$):

$$\left. \frac{dH}{dt} \right|_{\text{noise}} = -\frac{\epsilon_0^2}{2} \left(1 - \frac{1}{n} \right) \sum_{k=1}^n \frac{1}{P_k}$$

This is strictly negative: ϵ_0 noise always *decreases* entropy. For a peaked distribution (P concentrated on one category): $\sum_k P_k^{-1} \gg n^2$, so the decay is faster. For uniform P : $\sum_k P_k^{-1} = n^2$, giving the minimum decay rate.

2.1 Numerical Verification

Distribution	H	$\sum 1/P_k$	Predicted dH/dt	Simulated
Uniform ($P_k = 1/n$)	2.079	64.0	-0.00160	-0.00161
Peaked ($r = 0.415$)	1.689	97.8	-0.00244	-0.00291

The theory matches simulation to within 20% for peaked distributions (higher-order terms contribute).

3 Connection to H_{before}

The measured values from instrumented V20.3: $H_{\text{after}} = 1.816$, $H_{\text{before}} = 1.689$, $T_{\text{avg}} = 25.3$ sweeps. Required decay rate: $0.127/25.3 = 0.0050/\text{sweep}$.

Formal: H_{before} from Decay Mechanism

The decay rate at the time-averaged P distribution is:

$$\left. \frac{dH}{dt} \right|_{\text{avg}} = -\frac{\epsilon_0^2}{2} \left(1 - \frac{1}{n} \right) \mathbb{E} \left[\sum_k \frac{1}{P_k} \right]_{\text{events}}$$

Predicted at $r = 0.415$: 0.00244/sweep. Required: 0.0050/sweep. Factor-of-2 discrepancy indicates the effective peakedness at events is $r^* \approx 0.72$ ($H \approx 1.04$), i.e. nodes are more peaked than the mean at the moment they fire. This is consistent: high-KL nodes trigger more, and high-KL corresponds to more peaked P .

The mechanism is correct; the exact rate requires knowing $\mathbb{E}[\sum_k P_k^{-1}]$ at R-events.

4 The Bracket Identity: Final Boundary

With $f_{12} = d_s/2$ (derived) and the bracket decomposition:

$$\text{bracket} = H_{\text{after}} \frac{\kappa}{1+M} + \underbrace{(H_{\text{after}} - H_{\text{before}})}_{\delta H} \cdot \ln(1+M)$$

For $\text{bracket} = \kappa/\ln 2$:

$$\frac{H_{\text{after}}}{1+M} + \frac{\delta H}{\kappa} \ln(1+M) = \frac{1}{\ln 2}$$

Open: Bracket Identity Requires $\mathbb{E}[\sum_k P_k^{-1}]$ at Events

δH is determined by the decay rate dH/dt integrated over T_{avg} . From the derived ODE:

$$\delta H = T_{\text{avg}} \cdot \frac{\epsilon_0^2}{2} \left(1 - \frac{1}{n} \right) \mathbb{E} \left[\sum_k \frac{1}{P_k} \right]_{\text{events}}$$

The bracket identity $= \kappa/\ln 2$ is equivalent to:

$$T_{\text{avg}} \cdot \mathbb{E} \left[\sum_k \frac{1}{P_k} \right]_{\text{events}} = \frac{2n\kappa}{\epsilon_0^2(n-1) \ln(1+M) \ln 2} \left(\frac{1}{\ln 2} - \frac{H_{\text{after}}}{1+M} \right)^{-1}$$

This is a non-trivial identity relating the KL concentration of the P distribution at firing time to the geometric parameters M, κ, n, ϵ_0 .

Fastest path to close: instrument V20.3 to record $\sum_k P_{i,k}^{-1}$ at each R-event. If the resulting $T_{\text{avg}} \cdot \mathbb{E}[\sum_k P_k^{-1}]$ matches the RHS above, the bracket identity is numerically confirmed and the derivation reduces to explaining why this identity holds analytically.

5 Complete Status Table

Result	Status	Note
β formula at 1.1%	Closed	CRF_21a_Closure
$f_{12} = d_s/2 = 3/2$	Derived	Kuramoto; 1.3%
$H_{\text{after}} = \ln n - D_{\text{KL}}(\bar{P} U)$	Derived	Born rule; 0.9%
$\rho = 1 - \frac{1}{2}(n_d/N)^{1/d_s}$	Derived	Domain geometry
$dH/dt = -\frac{\epsilon_0^2}{2}(1 - \frac{1}{n}) \sum P_k^{-1}$	Derived	Taylor of H ; verified
Mechanism: $H_{\text{before}} < H_{\text{after}}$	Derived	From ϵ_0 decay
$r^* \approx 0.72$ at R-events	Hypothesis	Inferred from decay rate
bracket = 0.2197	Confirmed	Measured
bracket = $\kappa/\ln 2 = 0.2164$	Watching	1.5% gap
$\mathbb{E}[\sum P_k^{-1}]$ at events	Open	Key missing piece
$ \beta = d_s/(2 \ln 2)$	Open	#21; 2.2% gap

6 Roadmap

Path to Close #21

Next step (one simulation run): Add $\sum_k P_{i,k}^{-1}$ to the per-event log in the instrumented V20.3. From 2,500 events, compute: $T_{\text{avg}} \cdot \mathbb{E}[\sum P_k^{-1}]$ and check against the bracket-identity target. This measurement either confirms the bracket identity (numerically closing #21) or reveals the residual correction needed.

If confirmed: The bracket identity is numerically closed. The remaining work is an analytic mean-field theory explaining why $\mathbb{E}[\sum P_k^{-1}]$ takes the specific value it does—connecting the crystallisation dynamics to the information-theoretic constant $1/\ln 2$.

If not confirmed: The 1.5% bracket gap is real and the conjecture $|\beta| = d_s/(2 \ln 2)$ requires revision. This would be important negative information.

$\sum_k P_k^{-1}$ is the information content of the inverse distribution. When this quantity, averaged over R-events and weighted by time, equals $2n\kappa/[\epsilon_0^2(n-1)\text{bracket}]$, the bracket becomes $\kappa/\ln 2$ and the price of becoming actual in d_s dimensions is one dimension's worth of information measured in the universal unit $\ln 2$. One measurement settles this.